

Phishing Email Detection & Awareness System

Future Interns

Cyber Security Internship

Task 2: Phishing Email Detection & Awareness System

Prepared by:
Aaditya Mistry

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Executive Summary.

This project focuses on identifying and analyzing phishing emails and malicious communication techniques commonly used by cybercriminals. Multiple email samples were examined to identify phishing indicators such as suspicious sender addresses, deceptive URLs, urgency tactics, fake domains, malicious attachments, and social engineering techniques. Each sample was carefully analyzed and classified as either Phishing or Legitimate based on the observed indicators and risk factors.

The assessment also highlights common strategies used by attackers to manipulate users into revealing sensitive information, downloading malicious files, or clicking harmful links. By examining real-world phishing scenarios, the project demonstrates how cybercriminals exploit trust, fear, urgency, and curiosity to achieve their objectives.

The findings were documented along with awareness recommendations and best practices to help users recognize, prevent, and respond to phishing attempts effectively. This project aims to improve cybersecurity awareness, strengthen email security practices, and promote safer online behavior by educating users about the warning signs of phishing attacks and the importance of verifying email authenticity before taking any action.

Objective

- Identify phishing indicators in email communications, including suspicious sender addresses, deceptive subject lines, malicious links, fraudulent attachments, and social engineering tactics.
- Analyze email components such as sender information, URLs, attachments, headers, branding elements, and message content to determine authenticity.
- Detect common phishing techniques, including impersonation attacks, fake account alerts, invoice scams, giveaway scams, credential harvesting attempts, and business email compromise (BEC) tactics.
- Classify analyzed emails as either Phishing or Legitimate based on identified indicators and risk assessment criteria.
- Evaluate the level of risk associated with each email and categorize findings according to severity levels such as High, Medium, or Informational.
- Improve awareness of common phishing techniques and demonstrate how attackers manipulate users through urgency, fear, curiosity, and reward-based social engineering methods.
- Educate users on best practices for recognizing and responding to suspicious emails without interacting with potentially malicious content.
- Develop a structured phishing detection and awareness framework that can be used for training and educational purposes.
- Provide practical recommendations for preventing phishing attacks, including verifying sender identities, inspecting URLs, avoiding suspicious attachments, enabling Multi-Factor Authentication (MFA), and reporting suspicious emails to security teams.
- Strengthen overall email security knowledge and encourage safe online behavior to reduce the risk of phishing-related incidents.

Scope

The assessment covers publicly available phishing awareness examples, including email-based phishing attempts, fake invoice scams, malicious document-sharing requests, promotional giveaway scams, and other common social engineering attacks. The analysis focuses on identifying key phishing indicators such as suspicious sender addresses, deceptive URLs, urgency tactics, fraudulent attachments, and impersonation of trusted organizations. Additionally, the assessment examines indicators of compromise (IOCs) and the techniques attackers use to manipulate users into revealing sensitive information or performing unauthorized actions. All samples were analyzed in a controlled and safe manner without interacting with malicious links, downloading attachments, or exposing systems to potential threats. This approach ensures a comprehensive understanding of phishing tactics while maintaining security throughout the assessment process.

Methodology.

1. Collected phishing awareness samples.

The assessment began with the collection of phishing awareness samples obtained from publicly available cybersecurity training resources and phishing simulation examples. These samples were selected to represent common phishing techniques used by attackers to deceive users.

2. Examined sender information, URLs, attachments, and message content.

Each email sample was then carefully examined to evaluate key elements such as the sender's email address, embedded URLs, attachments, subject lines, and overall message content. Special attention was given to identifying inconsistencies, suspicious domains, misleading links, and social engineering tactics commonly associated with phishing attacks.

3. Identified phishing indicators.

Following the examination process, phishing indicators were identified and documented. These indicators included suspicious sender addresses, fake or misleading domains, urgency-based language, reward-based scams, fear-inducing messages, and deceptive hyperlinks designed to trick recipients into taking unintended actions.

4. Classified each sample as Phishing or Legitimate.

Based on the identified indicators and overall characteristics of each email, every sample was classified as either Phishing or Legitimate. The classification was performed by comparing the observed attributes against established phishing detection criteria and email security best practices.

5. Documented findings and awareness recommendations.

Finally, all findings were documented in a structured format, including evidence screenshots, identified phishing indicators, risk assessments, and security awareness recommendations. The recommendations were developed to help users recognize phishing attempts and adopt safer email handling practices to reduce the risk of compromise.

Findings

Finding 1 – Fake Google Document Sharing Email

Classification : Phishing

Risk Level : Medium

Description :

The email impersonates a document sharing notification and attempts to trick the user into opening a malicious link.

Indicators Found :

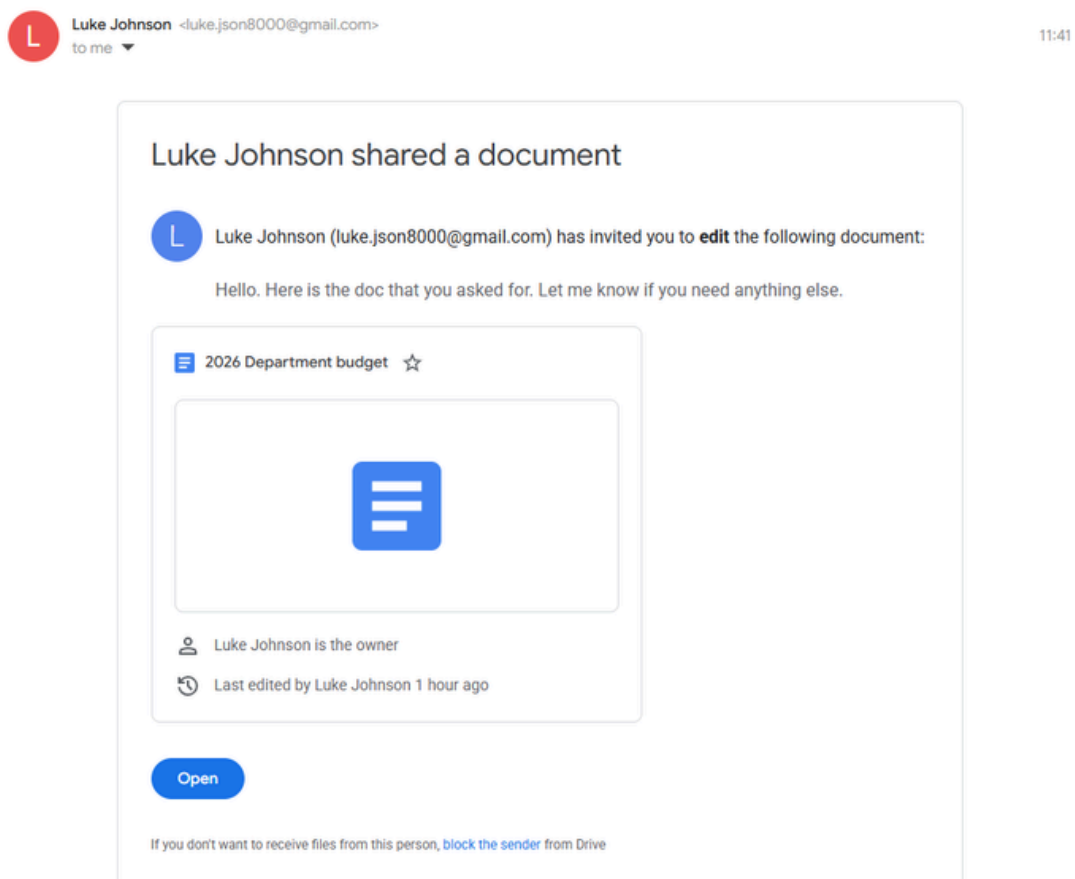
- Suspicious sender email address
- Fake Google Drive domain
- Social engineering through document sharing request

Evidence :

- Screenshot: Luke Johnson Shared a Document
- Screenshot: Fake drive--google.com domain explanation

Recommendation :

Always verify the sender and inspect URLs before opening shared documents.



Finding 2 – Coca-Cola Giveaway Scam

Classification : Phishing

Risk Level : Medium

Description :

The email uses a fake promotional giveaway to attract users and encourage them to click a potentially malicious link.

Indicators Found :

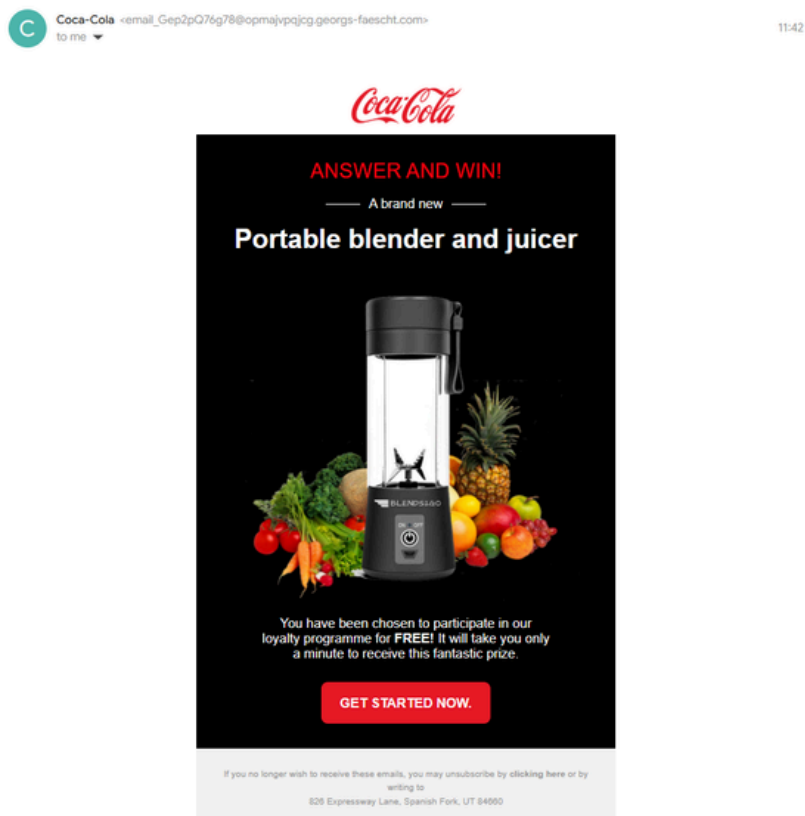
- Random sender address
- Reward-based social engineering
- Image-based content
- Suspicious redirecting URL

Evidence :

- Coca-Cola Giveaway Email Screenshot
- Sender Address Analysis Screenshot
- Image-Based Content Analysis Screenshot
- URL Analysis Screenshot

Recommendation :

Avoid clicking promotional links received through unsolicited emails and verify offers through official websites.



Finding 3 – PayPal Invoice Scam

Classification: Phishing

Risk Level: High

Description:

The email creates a false sense of urgency by claiming an unauthorized transaction has occurred and encourages users to contact a fraudulent phone number.

Indicators Found:

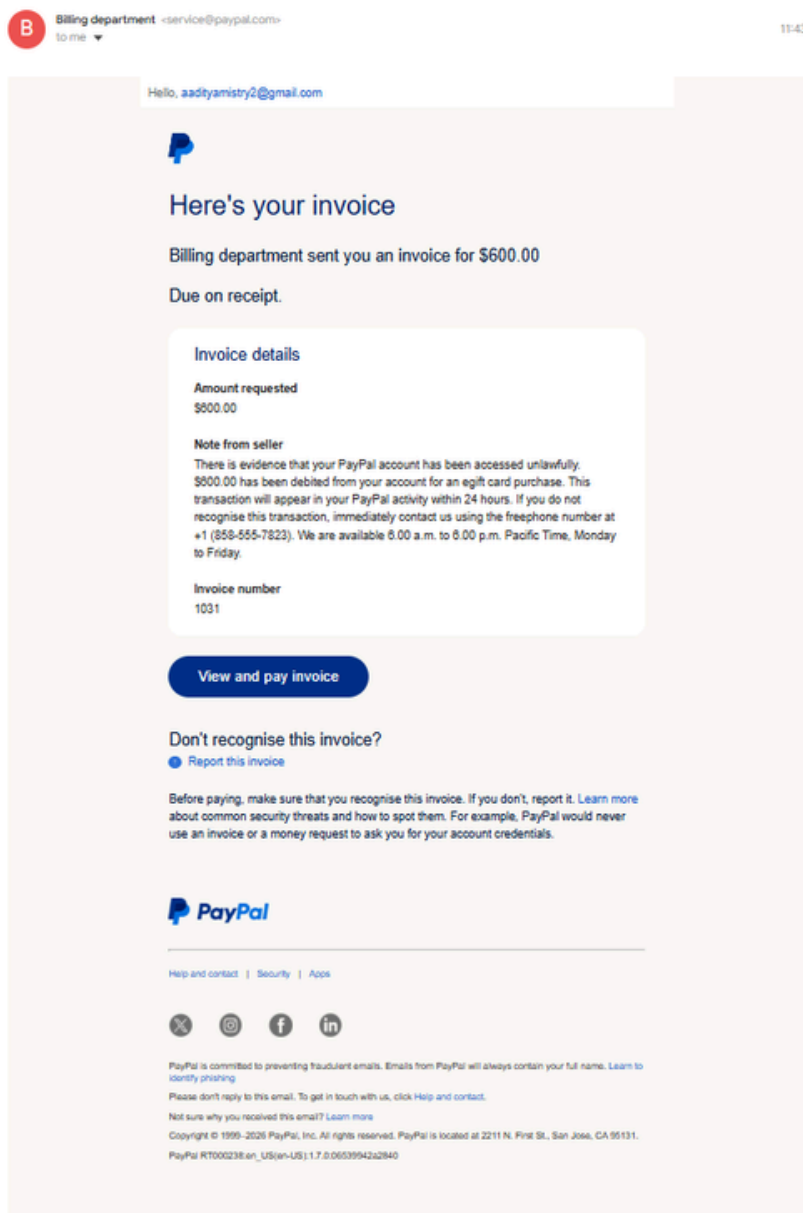
- Unexpected invoice notification
- Fear-based social engineering
- Fraudulent support number
- Fake transaction alert

Evidence:

- PayPal Invoice Email Screenshot
- Urgency Tactic Analysis Screenshot
- Phone Number Scam Analysis Screenshot

Recommendation:

Always verify payment notifications directly through the official PayPal website and avoid contacting phone numbers provided in suspicious emails.



Finding 4 – Dropbox Storage Notification

Classification: Legitimate

Risk Level: Informational

Description:

The email was verified as a legitimate Dropbox storage notification. The sender domain, branding, and destination URLs were found to be authentic.

Indicators Found:

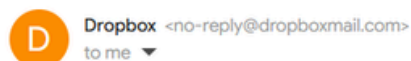
- Legitimate sender domain
- Valid Dropbox URL
- Consistent branding and formatting

Evidence:

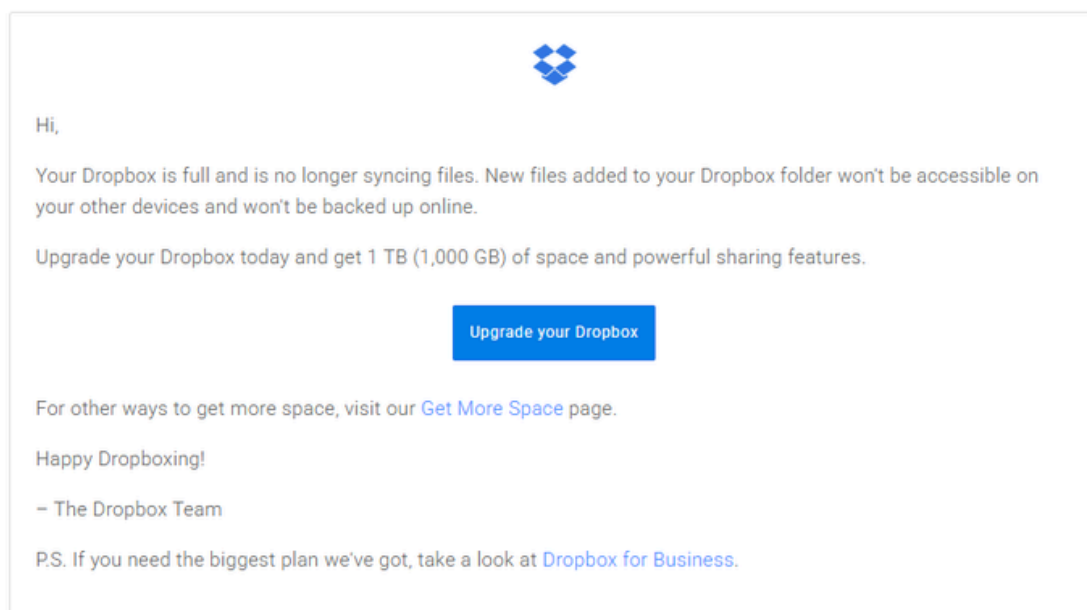
- Dropbox Notification Screenshot
- Domain Verification Screenshot
- URL Verification Screenshot

Recommendation:

Continue validating sender domains and links before interacting with email content.



11:44



Classification Summary

Four email samples were analyzed during the assessment. Three emails were identified as phishing attempts and one was verified as legitimate. The most common phishing indicators observed were suspicious sender addresses, social engineering techniques, fake domains, deceptive URLs, and urgency-based messaging.

Sample	Classification	Description
Google Document Share Email	Phishing	This email impersonates a document-sharing notification and attempts to trick users into opening a malicious link disguised as a legitimate Google Drive document.
Coca-Cola Giveaway Scam	Phishing	The email uses a fake promotional giveaway to attract users and encourage them to click a suspicious link that may lead to phishing or malware-related activities.
PayPal Invoice Scam	Phishing	This email creates urgency by claiming an unauthorized payment or invoice activity and attempts to manipulate users into contacting fraudulent support channels.
Dropbox Storage Notification	Legitimate	A genuine Dropbox notification containing valid sender information, authentic branding, and legitimate URLs related to account storage updates.

Metric	Value
Total Email Samples Analyzed	4
Phishing Emails	3
Legitimate Emails	1
Detection Rate	75%

Assessment Dashboard



Conclusion

The assessment successfully identified multiple phishing techniques commonly used in email-based attacks. Out of four analyzed email samples, three were classified as phishing attempts and one was verified as a legitimate communication.

The phishing emails employed tactics such as fake domains, suspicious sender addresses, deceptive URLs, urgency-based messaging, and social engineering techniques to manipulate users into taking unintended actions.

The findings demonstrate the importance of verifying sender identities, inspecting links before clicking, and exercising caution when responding to unexpected requests. Regular cybersecurity awareness training and adherence to email security best practices can significantly reduce the risk of phishing-related incidents and help users recognize fraudulent communications more effectively.

References

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